# Straight lines in the plane

## Lecture and worked examples

Maths Explorer

## Direction vector and Cartesian equation

A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs $(AB)$. Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

## Direction vector and Cartesian equation

For $A(-2, 4)$ and $B(6, 2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4, -1)$ directs $(AB)$.

## Worked answer: Direction vector and Cartesian equation

$\overrightarrow{AB} = (8, -2) = 2\vec{u}$. The vector $\vec{u}$ is nonzero and collinear with $\overrightarrow{AB}$; it therefore directs $(AB)$.

## Direction vector and Cartesian equation

The line through $A(x_A, y_A)$ with direction vector $\vec{u}(\alpha, \beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

## Direction vector and Cartesian equation

A point $M(x, y)$ belongs to the line if and only if $\overrightarrow{AM}$ is collinear with $\vec{u}$, including for $M = A$. Now

$$\det(\overrightarrow{AM}, \vec{u}) = (x - x_A)\beta - (y - y_A)\alpha.$$

Setting the determinant to zero gives the stated equation. Expanding, $a = \beta$, $b = -\alpha$ and $c = -\beta x_A + \alpha y_A$; since $\vec{u} \neq \vec{0}$, $(a, b) \neq (0, 0)$.

## Direction vector and Cartesian equation

The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

## Determining the equation of a line

From a point and a direction vector: choose $a, b$ then replace $x, y$ by the coordinates of the point to find $c$. From two distinct points: first calculate a direction vector.

## Determining the equation of a line

Determine a Cartesian equation of the line through $A(2, -1)$ with direction vector $\vec{u}(3, 2)$.

## Worked answer: Determining the equation of a line

$a = 2, b = -3$ will do. The line therefore has an equation $2x - 3y + c = 0$. Since $A$ belongs to it, $4 + 3 + c = 0$, so $c = -7$. An equation is $2x - 3y - 7 = 0$. Check:
$2 \times 2 - 3 \times (-1) - 7 = 0$ and $(3, 2)$ is indeed a direction vector.

## Determining the equation of a line

Determine a Cartesian equation of $(AB)$ for $A(3, 3)$ and $B(2, -2)$.

## Worked answer: Determining the equation of a line

$\overrightarrow{AB} = (-1, -5)$ directs the line. We can choose $(a, b) = (5, -1)$. With $A$, $15 - 3 + c = 0$, so $c = -12$. Thus $5x - y - 12 = 0$. For $B$, $10 + 2 - 12 = 0$, which confirms the result.

## Determining the equation of a line

Does the point $P(1, 2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2, 1)$ belong to it?

## Worked answer: Determining the equation of a line

For $P$, $2 - 6 + 4 = 0$: yes. For $Q$, $4 - 3 + 4 = 5 \neq 0$: no.

## Slope-intercept equation, slope and affine function

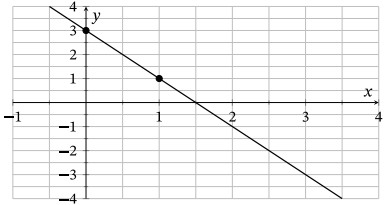
A non-vertical line has a unique slope-intercept equation $y = mx + p$. $m$ is the slope, $p$ the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1, m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of $x$. A horizontal line has an equation $y = k$.

## Slope-intercept equation, slope and affine function

If $x_A \neq x_B$, the slope of $(AB)$ is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find $p$, we then use $y_A = mx_A + p$.

# Slope-intercept equation, slope and affine function

Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.



## Worked answer: Slope-intercept equation, slope and affine function

$3y = -2x + 1$, so $y = -\frac{2}{3}x + \frac{1}{3}$. For $d$, plot $(0, 3)$ and $(1, 1)$, then draw the line; a move of 1 along the x-axis corresponds to $-2$ on the y-axis.

## Slope-intercept equation, slope and affine function

Find the affine function $f$ such that $f(0) = 3$ and $f(-1) = 5$, then the affine function $g$ whose graph passes through $C(-1, 2)$ and $E(5, -2)$.

## Worked answer: Slope-intercept equation, slope and affine function

$f(x) = -2x + 3$, since $m = (3 - 5)/(0 + 1) = -2$. For $g$, $m = (-2 - 2)/(5 + 1) = -2/3$ and $p = 2 - 2/3 = 4/3$, so $g(x) = -\frac{2}{3}x + \frac{4}{3}$.

## Relative positions of two lines

Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

## Relative positions of two lines

Determine an equation of $d$ through $A(2, 3)$ and parallel to $d' : x - 3y - 3 = 0$.

## Worked answer: Relative positions of two lines

$d'$ has direction vector $(3, 1)$. We look for $d : x - 3y + c = 0$. Since $A \in d$, $2 - 9 + c = 0$, so $c = 7$. Thus $d : x - 3y + 7 = 0$. The two lines are strictly parallel, because the constants differ for the same coefficients of $x, y$.

## Relative positions of two lines

Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

## Worked answer: Relative positions of two lines

1. The second equation is twice the first: coincident lines. 2. Two distinct vertical lines: strictly parallel. 3. The slopes 2 and $-1$ differ: intersecting. Their intersection satisfies $2x - 3 = -x + 3$, so $x = 2$ and $y = 1$.

## Intersection: solving by substitution

Solving a system of two equations in $x, y$ means finding all the pairs $(x, y)$ satisfying both equations simultaneously. For two equations of lines: intersecting lines give one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

## Intersection: solving by substitution

Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

# Intersection: solving by substitution

Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

## Worked answer: Intersection: solving by substitution

The first equation gives $y = 2x - 3$. Substituting into the second: $3x - 5(2x - 3) - 2 = 0$, that is $-7x + 13 = 0$. Thus $x = 13/7$ then $y = 5/7$. The solution set is $S = \{(13/7, 5/7)\}$. The substitutions give $(-26 + 5 + 21)/7 = 0$ and $(39 - 25 - 14)/7 = 0$.

# Intersection: solving by substitution

Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

## Worked answer: Intersection: solving by substitution

(a) $y = 2x - 3$ then $x - 4(2x - 3) + 5 = 0$, so $x = 17/7$, $y = 13/7$.

(b) $x = (10 - 3y)/2$; $2(10 - 3y) + 5y - 16 = 0$, so $y = 4$, then $x = -1$.

## Linear combination and Python

Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

## Linear combination and Python

Solve $\begin{cases} 2x + 3y + 1 = 0 \\ 3x - 4y - 2 = 0. \end{cases}$ Then redo the two systems of the previous example by linear combination.

## Worked answer: Linear combination and Python

Multiplying the first equation by 3 and the second by $-2$ gives $6x + 9y + 3 = 0$ and $-6x + 8y + 4 = 0$. The sum gives $17y + 7 = 0$, so $y = -7/17$. Then $2x - 21/17 + 1 = 0$, so $x = 2/17$. For (a), add four times the first equation to the second: $-7x + 17 = 0$, so $(x, y) = (17/7, 13/7)$. For (b), subtract twice the first from the second: $-y + 4 = 0$, so $(x, y) = (-1, 4)$.

# Python

```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

## Linear combination and Python

Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

## Worked answer: Linear combination and Python

`True`, then `False`: the determinants are 0 and 2. The exact test is suitable here for integer coordinates; floating-point approximations require taking numerical errors into account.